PSEUDOCODE

JAVA PROGRAMMING

NAME:PRETHIVIKA G

REG NO:192521085

DEPT: B.Tech Information Technology

QUESTION 1:

Q1 **Student Management System [BL3 – Apply]** 10 marks

Write pseudocode to design a Student class with attributes (name, rollNo, marks) and methods to:

- Accept student details
- Calculate average marks
- Display result (Pass/Fail)

Apply encapsulation and data hiding by restricting direct access to attributes.

OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

Your Answer

```
START
CLASS Student
 PRIVATE
   name
   rollno
   marks
   avg
 PUBLIC
 METHOD input()
    INPUT name
    INPUT rollno
    INPUT n
    FOR(i=0 to n)
        INPUT marks[i]
    END FOR
 END METHOD
 METHOD calavg()
   sum=0
   FOR(i=0 to n)
     sum=sum+marks[i]
   END FOR
   avg=sum/n
 END METHOD
 METHOD display()
  PRINT "Student name",name
  PRINT "Roll number",rollno
  PRINT "Average",avg
  IF avg>=50
     PRINT "Result : Pass"
  ELSE
     PRINT "Result :Fail"
  END IF
 END METHOD
END CLASS
BEGIN MAIN
  CREATE std AS Student
  CALL std.input()
  CALL std.calavg()
  CALL std.display()
END MAIN
END
```

QUESTION 2:

Q2 **Library Book Management [BL3 – Apply]** 10 marks

Design a Book class with private attributes (title, author, ISBN, isAvailable). Implement methods to:

- Issue a book (change availability status)
- Return a book
- Display book details using getter methods

Demonstrate object creation for at least 3 books and simulate issue/return operations.

OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer

```
START
CLASS book
 PRIVATE
   title
   author
   ISBN
   isavail
 PUBLIC
   METHOD setdet(t,a,i)
     title=t
     author=a
     ISBN=i
     isavail=TRUE
   END METHOD
```

```
  METHOD getISBN()
      RETURN ISBN
  END METHOD
  METHOD getavail()
      RETURN isavail
  END METHOD
  METHOD issuebook()
      IF isavail=TRUE THEN
          isavail=FALSE
          PRINT "book issued successfully"
      ELSE
          PRINT "book is already issued"
      END IF
  END METHOD
  METHOD returnbook()
     IF isavail=FALSE THEN
          isavail=FALSE
          PRINT "book returned successfully"
     ELSE
          PRINT" book is already available"
     END IF
  END METHOD
METHOD displaybook()
     DISPLAY "TITLE: ",gettitle()
     DISPLAY "AUTHOR: ",getauthor()
```

```
        isavail=FALSE
        PRINT "book issued successfully"
      ELSE
        PRINT "book is already issued"
      END IF
  END METHOD
  METHOD returnbook()
    IF isavail=FALSE THEN
      isavail=FALSE
      PRINT "book returned successfully"
    ELSE
      PRINT" book is already available"
    END IF
  END METHOD
METHOD displaybook()
    DISPLAY "TITLE: ",gettitle()
    DISPLAY "AUTHOR: ",getauthor()
    DISPLAY "ISBN: ",getISBN()
    DISPLAY "availability: ",getavail()
  END METHOD
END CLASS
```

QUESTION 3:

Q3 **Employee Salary Calculation [BL3 – Apply]** 10 marks

Write pseudocode for an Employee class with encapsulated attributes (empId, name, basicPay, designation). Implement methods to: • Calculate gross salary (basic + HRA + DA)
• Calculate net salary after deductions (PF, tax)
• Display pay slip
Apply data hiding so salary components are not directly accessible.
OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

```
START
CLASS employee
PRIVATE
  empid
  name
  basicpay
  desig
  hra
  da
  pf
  tax
  grosssal
  net sal
PUBLIC
```

```
PUBLIC
  METHOD setemp(id,n,pay,des)
     empid=id
     name=n
     basicpay=pay
     desig=des
  END METHOD
  METHOD calgross()
     hra=basicpay*20/100
     da=basicpay*10/100
     grossal=basicpay+hra+da
  END METHOD
  METHOD calnet()
     pf=basicpay*12/100
     tax=grosssal*5/100
     netsal=grosssal-(pf+tax)
  END METHOD
  METHOD display()
      DISPLAY "emp id: ",empid
      DISPLAY " emp name: ",name
      DISPLAY "designation: ",desig
      DISPLAY "basic pay:",basicpay
      DISPLAY "gross salary: ",grosssal
      DISPLAY "net salary: ",netsal
  END METHOD
END CLASS
```

QUESTION 4:

Q4 **Shape Area Calculator using Polymorphism [BL3 – Apply]** 10 marks

Create a Shape base class with an overridable calculateArea() method. Apply this to:
- Circle — area using radius
- Rectangle — area using length and breadth
- Triangle — area using base and height

Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.
OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

```
START
CLASS shape
PUBLIC
METHOD calarea()
  DISPLAY "area cannot be calculated"
END METHOD
END CLASS
CLASS circle EXTENDS shape
PRIVATE
  rad
PUBLIC
METHOD setrad(r)
  rad=r
END METHOD
METHOD calarea()
  area=3.15*rad*rad
  DISPLAY "area of circle: ",area
END METHOD
END CLASS
CLASS rectangle EXTENDS shape
PRIVATE
  len
  bre
PUBLIC
METHOD setdim(l,b)
  len=l
  bre=b
end method
METHOD calarea()
  area=len*bre
  DISPLAY "area of rectangle: ",area
END METHOD
END CLASS
CLASS triangle EXTENDS shape
PRIVATE
  ba
  he
PUBLIC
```

```
PUBLIC
METHOD setdim(b,h)
  ba=b
  he=h
END METHOD
METHOD calarea()
  area=(ba*he)/2
  DISPLAY "area of triangle: ",area
END METHOD
END CLASS
BEGIN MAIN
CREATE c as circle
CREATE r as rectangle
CREATE  t as triangle
CALL c.setrad(7)
CALL r.setdim(10,5)
CALL t.setdim(8,6)
CALL c.calarea()
CALL r.calarea()
CALL t.calarea()
END MAIN
END
```

QUESTION 5:

Q5 **5. Vehicle Registration System [BL3 – Apply]** 10 marks

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:
• TwoWheeler — with engine capacity
• FourWheeler — with seating capacity and AC status
Apply inheritance to reuse common attributes and override a displayDetails() method in each subclass.
OOP Concepts: Inheritance | Method Overriding | Subclass

Your Answer

```
BEGIN
CLASS vehicle
PROTECTED
  reg
  own
  fuel
PUBLIC
METHOD setvehicle(r,o,f)
  reg=r
  own=o
  fuel=f
END METHOD
METHOD display()
  DISPLAY "registration no: ",reg
  DISPLAY "owner: ",owner
  DISPLAY "fuel type: ",fuel
END METHOD
END CLASS
CLASS twowheel EXTENDS vehicle
PRIVATE
  engcap
PUBLIC
METHOD seteng(cap)
  engcap=cap
END METHOD
METHOD display()
  DISPLAY "registration no: ",reg
  DISPLAY"owner: ",owner
  DISPLAY "fuel type: ",fuel
  DISPLAY "engine capacity: ",engcap
END METHOD
END CLASS
CLASS fourwheel EXTENDS vehicle
PRIVATE
  seat
  ac
PUBLIC
METHOD setfour(se,acc)
  seat=se
  ac=acc
```

```
  seat=se
  ac=acc
END METHOD
METHOD display()
  DISPLAY "registration no: ",reg
  DISPLAY "owner: ",owner
  DISPLAY  "fuel:",fuel
  DISPLAY "seating capacity ",seat
  DISPLAY "ac status",ac
END METHOD
END CLASS
BEGIN MAIN
CREATE
CREATE
CALL b.setvehicle("TN01AB","Rahul","petrol")
CALL b.seteng(150)
CALL c.setvehicle("TN09CD","OVIYA","diesel")
CALL c.setfour(5,"yes")
CALL b.display()
CALL c.display()
END MAIN
END
```

QUESTION 6:

Q6 6. Bank Account Hierarchy [BL4 – Analyze] 10 marks

Develop pseudocode for a BankAccount superclass and two subclasses — SavingsAccount (with interest calculation) and CurrentAccount (with overdraft facility). Analyze and implement:

- Inheritance of common attributes (accNo, balance, holder)
- Method overriding for withdrawal behavior
- How overdraft protection differs from savings limit

Compare the behavior of withdraw() across both subclasses and justify the design choices.

OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

```
BEGIN
CLASS bankacc
PROTECTED
  accno,holder,bal
PUBLIC
METHOD setdet(a,h,b)
  accno=a
  holder=h
  bal=b
END METHOD
METHOD Deposit(amt)
  bal=bal+amt
END METHOD
METHOD withdraw(amt)
  DISPLAY " withdrawal"
END METHOD
METHOD display()
  DISPLAY "acc num",accno
  DISPLAY "Holder",holder
  DISPLAY "balance",bal
END METHOD
END CLASS
CLASS saving EXTENDS bankacc
PRIVATE
 rate
PUBLIC
METHOD setint(r)
  rate=r
END METHOD
METHOD calint()
  inrt=bal+rate/100
  bal=bal+inrt
END METHOD
METHOD withdraw(amt)
  IF amt<=bal THEN
    bal=bal-amt
    DISPLAY "withdrawal successful"
  ELSE
    DISPLAY "insufficient balance"
```

```
END METHOD
END CLASS
CLASS curacc EXTENDS bankacc
PRIVATE
 ovl
PUBLIC
METHOD  withdraw(amt)
  IF amt<=(bal+ovl) THEN
    bal=bal-amt
    DISPLAY "withdrawal successful"
  ELSE
    DISPLAY "overdraft limit exceeded"
  END IF
END MEETHOD
END CLASS
BEGIN MAIN
CREATE s as saving
CREATE c as curacc
CALL s.setdet(1001,"RAHUL",20000)
CALL s.setint(5)
CALL c.setdet(2001,"priya",10000)
CALL c.setint(5000)
CALL s.calint()
CALL s.withdraw(5000)
CALL c.withdraw(12000)
CALL s.display()
```

QUESTION 7:

Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks

Design a Patient base class and analyze how it extends into:
• InPatient — with ward number, admission date, and daily charges
• OutPatient — with appointment date and consultation fee
Analyze encapsulation requirements for sensitive data (diagnosis, prescription). Override a generateBill() method for each type and compare the billing logic differences. OOP Concepts: Encapsulation | Inheritance | Polymorphism

Your Answer

```
BEGIN
CLASS patient
PROTECTED
  pid,name
PRIVATE
  dig,pres
PUBLIC
METHOD setdet(id,n,d,p)
  pid=id
  name=n
  dig=d
  pres=p
END METHOD
METHOD genbill()
END METHOD
```

```
END CLASS
CLASS inpatient Extends patient
PRIVATE
  ward,admnd,dch,day
PUBLIC
METHOD setinpatient(w,a,c,d)
  ward=w
  admnd=a
  dch=c
  day=d
END METHOD
METHOD genbill()
  bill=dch*day
  DISPLAY "inpatient bill: ",bill
END METHOD
END CLASS
CLASS outpatient EXTENDS patient
PRIVATE
  appda,cfee
PUBLIC
METHOD setoutpatient(a,f)
  appda=d
  cfee=f
END METHOD
METHOD genbill()
  bill=cfee
```

```
METHOD setoutpatient(a,f)
  appda=d
  cfee=f
END METHOD
METHOD genbill()
  bill=cfee
  DISPLAY "outpatient bill: ",bill
END METHOD
END CLASS
BEGIN MAIN
CREATE ip as inpatient
CREATE op as outpatient
CALL ip.setdet(101,"rahul","fever","medicine")
CALL ip.setinpatient(12,"10/07/2026",2000,5)
CALL op.setdet(102,"priya","cold","tablets")
CALL op.setoutpatient("12/07/2026",500)
CALL ip.genbill()
CALL op.genbill()
END MAIN
END
```

QUESTION 8:

Q8 **8. E-Commerce Product Catalog [BL4 – Analyze]** 10 marks

Analyze and design a Product class hierarchy for an e-commerce system:

- Electronics — with warranty, brand, voltage
- Clothing — with size, fabric, gender
- Grocery — with expiryDate and weight

Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts: Polymorphism | Inheritance | Method Overriding

Your Answer

```
  gen=g
END METHOD
METHOD applydis()
  price=price-(price*20/100)
  DISPLAY "clothing price",price
END METHOD
END CLASS
CLASS grocery EXTENDS product
PRIVATE
  expd,wei
PUBLIC
METHOD setgro(e,w)
  expd=e
  wei=w
END METHOD
```

```
END METHOD
METHOD applydis()
  price=price-(price*5/100)
  DISPLAY "grocery price",price
END METHOD
END CLASS
BEGIN MAIN
CREATE e as elect
CREATE c as clothing
CREATE g as grocery
CALL e.setdet(101,"laptop",60000)
CALL e.setelect(2,"dell",230)
CALL c.setdet(102,"shirt",1500)
CALL c.setcloth("l","cotton","men")
CALL g.setdet(103,"rice",1200)
CALL g.setgro("15/08/2026","10")
CALL e.applydis()
CALL c.applydis()
CALL g.applydis()
END MAIN
END
```

QUESTION 9:

Q9 9. University Staff Hierarchy [BL4 – Analyze] 10 marks

Analyze the design of a Staff base class extended by TeachingStaff and NonTeachingStaff. Further extend TeachingStaff into Professor and Lecturer (multilevel inheritance). For each level:

- Identify which attributes should be private vs protected
- Override calculateAllowance() at each level
- Analyze how protected access enables inheritance without full exposure Justify the use of multilevel inheritance and explain how access modifiers enforce data hiding.

OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding

Your Answer

```
BEGIN
CLASS staff
PROTECTED
  sid,name,dept
PRIVATE
  sal
PUBLIC
METHOD setdet(id,n,d,s)
  sid=id
  name=n
  dept=d
  sal=s
END METHOD
```

```
METHOD calall()
  DISPLAY "base allowance"
END METHOD
END CLASS
CLASS teach EXTENDS staff
PROTECTED
  sub
PUBLIC
METHOD calall()
  DISPLAY "teaching allowance=5000"
END METHOD
END CLASS
CLASS proff EXTENDS teach
PRIVATE
  resall
PUBLIC
METHOD calall()
  DISPLAY "professor allowance=1000"
END METHOD
END CLASS
CLASS lecturer EXTENDS teach
PRIVATE
  lehr
PUBLIC
METHOD calall()
  DISPLAY " lecturer allowance=7000"
```

```
  DISPLAY " lecturer allowance=7000"
END METHOD
END CLASS
CLASS nonteach EXTENDS staff
PRIVATE
  des
PUBLIC
METHOD calall()
  DISPLAY "non-teaching allowance=4000"
END METHOD
END CLASS
BEGIN CLASS
CREATE p as proff
CREATE l as lecturer
CREATE n as nonteach
CALL p.setdet(101."rahul","cse",80000)
CALL l.setdet(102,"priya","it",50000)
CALL n.setdet(103,"arun","admin",40000)
CALL p.calall()
CALL l.calall()
CALL n.calall()
END MAIN
END
```

QUESTION 10:

Q10 **10. Smart Home Device Controller [BL4 – Analyze]** 10 marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight, SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:

- How polymorphism allows a single controller loop to manage all devices
- Security implications of data hiding in SmartLock (PIN must never be exposed) Design the class hierarchy and explain the OOP principles applied at each level.

OOP Concepts: Polymorphism | Encapsulation | Inheritance

Your Answer

```
BEGIN
CLASS smartdev
PROTECTED
 devid
 devname
PUBLIC
METHOD setdet(id,name)
 devid=id
 devname=name
END METHOD
METHOD execom(cmd)
END METHOD
END CLASS
```

```
END CLASS
CLASS smlight EXTENDS smartdev
PUBLIC
METHOD execom(cmd)
 IF cmd="on" THEN
   DISPLAY "light turned on"
 ELSE IF cmd="off" THEN
   DISPLAY "light turned off"
 ELSE
   DISPLAY "light status displayed"
 END IF
END METHOD
END CLASS
CLASS smarttherm EXTENDS smartdev
PUBLIC
METHOD execom(cmd)
 IF cmd="on" THEN
   DISPLAY "thermostat activated"
 ELSE IF cmd="off" THEN
   DISPLAY "thermostat deactivated"
 ELSE
   DISPLAY "temperature status displayed"
 END IF
END METHOD
END CLASS
CLASS smartlock EXTENDS smartdev
```

```
PRIVATE
  pin
PUBLIC
METHOD setpin(p)
  pin=p
END METHOD
METHOD execom(cmd)
  IF cmd="on" THEN
    DISPLAY "door locked"
  ELSE IF cmd="off" THEN
    DISPLAY "door unlocked"
  ELSE
    DISPLAY "lock status dispalyed"
  END IF
END METHOD
END CLASS
BEGIN MAIN
CREATE li as smlight
CREATE th as smarttherm
CREATE lo as smartlock
CALL li.setdet(101,"bedroom light")
CALL th.setdet(102,"hall thermostat")
CALL lo.setdet(103,"main door lock")
CALL lo.setpin("1234")
CALL li.execom("on")
CALL th.execom("status")
```

```
  ELSE
    DISPLAY "lock status dispalyed"
  END IF
END METHOD
END CLASS
BEGIN MAIN
CREATE li as smlight
CREATE th as smarttherm
CREATE lo as smartlock
CALL li.setdet(101,"bedroom light")
CALL th.setdet(102,"hall thermostat")
CALL lo.setdet(103,"main door lock")
CALL lo.setpin("1234")
CALL li.execom("on")
CALL th.execom("status")
CALL lo.execom("off")
END MAIN
END
```